

Lab - PowerShell - Azure Web Apps

To Begin with the Lab

Summary

This lab demonstrates how to use **Azure PowerShell** to create and manage an Azure Web App. You will create a resource group, App Service Plan, Web App, deploy a sample application, verify the deployment, and remove resources after completing the lab.

Understanding

Azure PowerShell is a command-line tool used to automate Azure resource management. It provides PowerShell cmdlets for creating, configuring, deploying, and managing Azure services. Using PowerShell reduces manual work, ensures consistency, and is ideal for automation scripts.

Example:

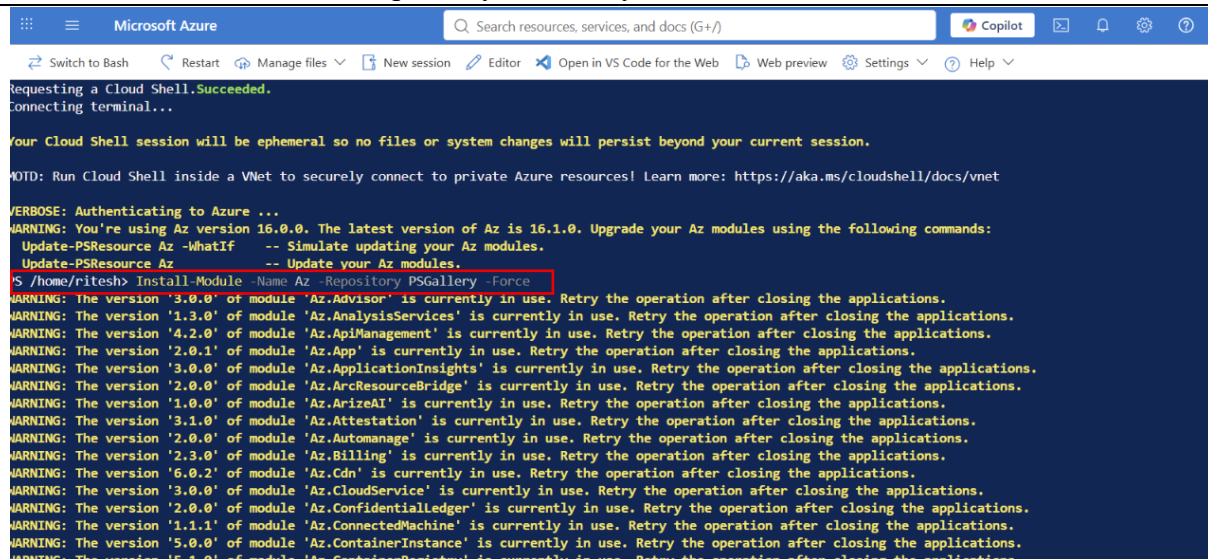
A company needs to deploy multiple web applications for different departments. Instead of creating each resource manually through the Azure Portal, an administrator uses Azure PowerShell scripts to automatically create Resource Groups, App Service Plans, and Web Apps, making deployment faster and more reliable.

Lab Steps

Step 1: Install Azure PowerShell

- Install the latest Azure PowerShell module.
- Open Windows PowerShell as Administrator.
- Run this command in **Azure power shell**.

Install-Module -Name Az -Repository PSGallery -Force



```
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Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.

NOTE: Run Cloud Shell inside a VNet to securely connect to private Azure resources! Learn more: https://aka.ms/cloudshell/docs/vnet

VERBOSE: Authenticating to Azure ...
WARNING: You're using Az version 16.0.0. The latest version of Az is 16.1.0. Upgrade your Az modules using the following commands:
  Update-PSResource Az -WhatIf -- Simulate updating your Az modules.
  Update-PSResource Az -- Update your Az modules.
$ /home/ritesh> Install-Module -Name Az -Repository PSGallery -Force
WARNING: The version '3.0.0' of module 'Az.Advisor' is currently in use. Retry the operation after closing the applications.
WARNING: The version '1.3.0' of module 'Az.AnalysisServices' is currently in use. Retry the operation after closing the applications.
WARNING: The version '4.2.0' of module 'Az.ApiManagement' is currently in use. Retry the operation after closing the applications.
WARNING: The version '2.0.1' of module 'Az.App' is currently in use. Retry the operation after closing the applications.
WARNING: The version '3.0.0' of module 'Az.ApplicationInsights' is currently in use. Retry the operation after closing the applications.
WARNING: The version '2.0.0' of module 'Az.ArcResourceBridge' is currently in use. Retry the operation after closing the applications.
WARNING: The version '1.0.0' of module 'Az.ArcizeAI' is currently in use. Retry the operation after closing the applications.
WARNING: The version '3.1.0' of module 'Az.Attestation' is currently in use. Retry the operation after closing the applications.
WARNING: The version '2.0.0' of module 'Az.Automation' is currently in use. Retry the operation after closing the applications.
WARNING: The version '2.3.0' of module 'Az.Billing' is currently in use. Retry the operation after closing the applications.
WARNING: The version '6.0.2' of module 'Az.Cdn' is currently in use. Retry the operation after closing the applications.
WARNING: The version '3.0.0' of module 'Az.CloudService' is currently in use. Retry the operation after closing the applications.
WARNING: The version '2.0.0' of module 'Az.Coverage' is currently in use. Retry the operation after closing the applications.
WARNING: The version '1.1.1' of module 'Az.ConnectedMachine' is currently in use. Retry the operation after closing the applications.
WARNING: The version '5.0.0' of module 'Az.ContainerInstance' is currently in use. Retry the operation after closing the applications.
WARNING: The version '5.1.0' of module 'Az.ContainerRegistry' is currently in use. Retry the operation after closing the applications.
```

- Verify the **installation**.
- Run this Command.

Get-Module -ListAvailable Az

```
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PS /home/ritesh> Get-Module -ListAvailable Az

Directory: /home/ritesh/.local/share/powershell/Modules

ModuleType Version PreRelease Name PSEdition ExportedCommands
-----
Script 16.1.0 Az Core,Desk

Directory: /usr/local/share/powershell/Modules

ModuleType Version PreRelease Name PSEdition ExportedCommands
-----
Script 16.0.0 Az Core,Desk

PS /home/ritesh>
```

Step 2: Sign in to Azure

- Run the Azure sign-in command.
- Authenticate using your Azure account.
- Select the required subscription if multiple **subscriptions** exist.
- Run this command.
- Click **Enter**

Set-AzContext -SubscriptionName "....."

```
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PS /home/ritesh> Set-AzContext -SubscriptionName "Visual Studio Enterprise Subscription"

Tenant: 3d18f3ae-8875-4771-aaa9-a9afcd43d751

SubscriptionName SubscriptionId Account Environment
-----
Visual Studio Enterprise Subscription 294539c9-c109-4db8-b785-0043716c0b11 MSI@50342 AzureCloud

PS /home/ritesh>
```

Step 3: Create a Resource Group

- Create a **new Resource Group**.
- Choose a **unique** Resource Group name.
- Select the desired **Azure region**.
- Click **Enter**

New-AzResourceGroup -Name **Webapp** -Location "East US"

```
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PS /home/ritesh> New-AzResourceGroup -Name Webapp -Location "East US"

ResourceGroupName : Webapp
Location           : eastus
ProvisioningState   : Succeeded
Tags               :
ResourceId          : /subscriptions/294539c9-c109-4db8-b785-0043716c0b11/resourceGroups/Webapp

PS /home/ritesh>
```

Step 4: Create an App Service Plan

- Create an App Service Plan.

- Enter the Name **WebAppServicePlan**.
- Enter the Location **East US**.
- Enter the ResourceGroupName **Webapp**.
- Select the pricing tier **Free**.
- Run this Command

```
New-AzAppServicePlan `
-Name WebAppServicePlan `
-Location "East US" `
-ResourceGroupName Webapp `
-Tier Free
```

- Click **Enter**

```
PS /home/ritesh> New-AzAppServicePlan `
>> -Name WebAppServicePlan `
>> -Location "East US" `
>> -ResourceGroupName Webapp
>> Tier Free
```

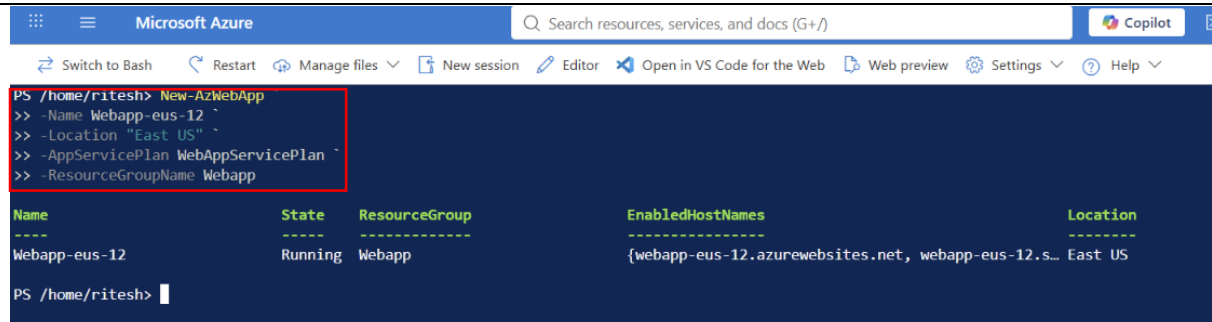
Name	State	ResourceGroup	EnabledHostNames	Location
WebAppServicePlan		Webapp		East US

```
PS /home/ritesh> |
```

Step 5: Create an Azure Web App

- Create a Web App.
- Provide a globally unique Web App name.
- Associate it with the App Service Plan.
- Run this command.

```
New-AzWebApp `
-Name Webapp-eus-12 `
-Location "East US" `
-AppServicePlan WebAppServicePlan `
-ResourceGroupName Webapp
```



```
PS /home/ritesh> New-AzWebApp
>> -Name Webapp-eus-12 `
>> -Location "East US" `
>> -AppServicePlan WebAppServicePlan `
>> -ResourceGroupName Webapp
```

Name	State	ResourceGroup	EnabledHostNames	Location
Webapp-eus-12	Running	Webapp	{webapp-eus-12.azurewebsites.net, webapp-eus-12.s...	East US

```
PS /home/ritesh> |
```

- Properties command for AzResource

```
Set-AzResource `
-ResourceGroupName "Webapp" `
-ResourceType "Microsoft.Web/sites" `
-ResourceName "Webapp-eus-12" `
-Properties @{ linuxFxVersion = "DoTNETCORE|10.0" } `
-Force
```

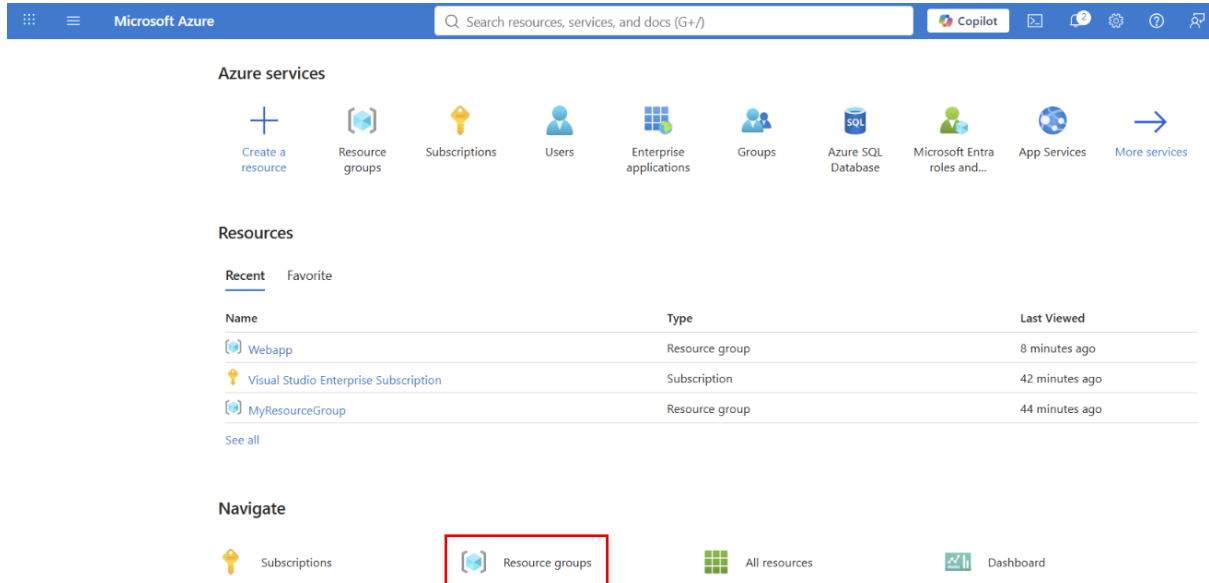
```
PS /home/ritesh> Set-AzResource `
>> -ResourceGroupName "Webapp" `
>> -ResourceType "Microsoft.Web/sites" `
>> -ResourceName "Webapp-eus-12" `
>> -Properties @{ linuxFxVersion = "DOTNETCORE|10.0" } `
>> -Force
```

SiteName	Location	State	OutboundIpAddresses	EnabledHostInfo	WebSpace
Webapp-eus-12	East US	Running	135.237.48.119 20.119.8.0 20.119.8.101	webapp-eus-12.azurewebsites.net* SSL=Disabled webapp-eus-12.scm.azurewebsites.net SSL=Disabled	Webapp-EastUSWebSpace

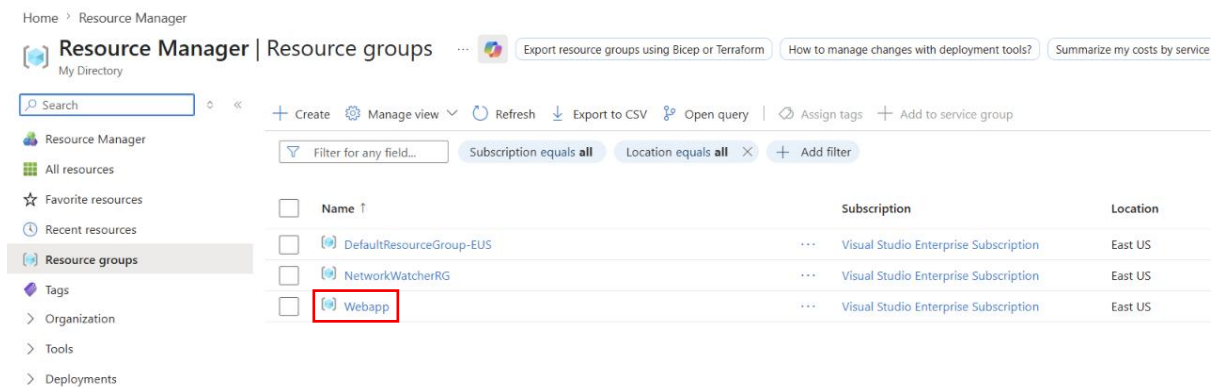
```
PS /home/ritesh>
```

Step 6: Verify the Web App

- Go to the Azure Resource Group.



- Navigate to **Webapp** Resource Group.



- App service and Webapp is Created Successfully

[+ Create](#) [Manage view](#) [Delete resource group](#) [Refresh](#) [Export to CSV](#) [Open query](#) [Assign tags](#)

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

Essentials

Resources Recommendations

Type equals all Location equals all [+ Add filter](#)

☐	Name ↑	Type	Location
☐	 Webapp-eus-12	...	App Service East US
☐	 WebAppServicePlan	...	App Service plan East US